

to absorb heat. When you pour water on your skin, it makes you feel cool. As water evaporates, it absorbs heat, creating a cool feeling. Rubbing alcohol makes you feel even cooler because it evaporates at a lower temperature. The liquid refrigerant used in a refrigerator evaporates at an extremely low temperature, creating the freezing temperatures inside the refrigerator.

There are four basic parts to any refrigerator—compressor, heat-exchanging pipes, expansion valve and refrigerant. The basic mechanism of a refrigerator is fairly standard. The compressor compresses the refrigerant gas, which raises the refrigerant's pressure and temperature, so the heat-exchanging coils outside the refrigerator allow the refrigerant to dissipate the heat of pressurisation. As it cools, the refrigerant condenses into liquid form and flows through the expansion valve. At this stage, while the liquid refrigerant moves from a high-pressure zone to a low-pressure zone, it expands and evaporates. While evaporating, it absorbs heat, making the space cold. The coils inside the refrigerator allow the refrigerant to absorb heat, making the inside of the refrigerator cold. This cycle repeats.

Conventional cooling systems of refrigerators or air-conditioners require energy-eating compressors and lots of heating coils, which had been developed years ago without any further improvement in coefficient of performance (COP). They rely on the properties of gases to cool and most systems use the change in density of gases at changing pressures to cool. The coolants commonly used are either harmful to people or the environment.

In the future, it is possible that the cooling power of a noisy, bulky household refrigerator will be available on a small, lightweight device without any moving part. The same device, when given a heat source like a car's exhaust pipe, could be used to generate electricity.

Efforts are underway to develop other types of refrigeration technology

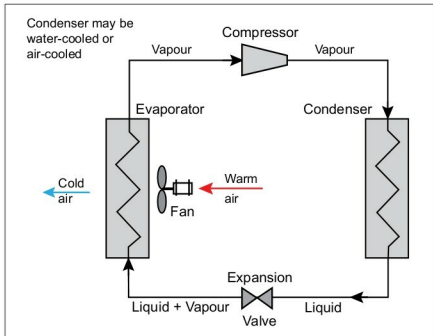


Fig. 1: Typical single-stage vapour compression refrigeration

gies like electric refrigeration, magnetic refrigeration, thermoelectric refrigeration and thermoacoustic refrigeration, which will be eco-friendly, cost-effective and efficient.

Electric refrigeration: Without coils and compressors

Efforts are being made to develop an electric field refrigeration unit (as electricity is more convenient) and in the future, we can envision a flat-panel refrigerator with no coils and compressors, but solid polymer with appropriate heat exchangers.

Researchers are working with ferroelectric polymers that exhibit temperature changes at room temperature under an electrical field. These polar polymers include poly(vinylidene fluoride-trifluoroethylene) and poly(vinylidene fluoride-trifluoroethylene)-chlorofluoroethylene. However, there are other polar polymers that exhibit the same effect.

The approach uses the change from disorganised to organised state that occurs in some polar polymers when placed in an electric field. The natural state of these materials is disorganised

with the various molecules randomly positioned. When electricity is applied, the molecules become highly ordered and the material gives off heat and becomes colder. When electricity is turned off, the material reverts to its disordered state and absorbs heat. Researchers report a change in temperature of about -5°C (22.6°F) in the material.

Repeated randomisation and ordering of the material combined with an appropriate heat exchanger could provide a wide range of heating and cooling temperatures. The polymers are flexible and can be used for heating and cooling, so there may be many different possible applications.

Magnetic refrigeration: Milestone in cooling

Researchers are also attempting to make magnetic fields do the cooling. Magnetic cooling technology exploits the fact that when a magnetic material (like gadolinium) is magnetised, heat is produced as a byproduct of entropy. The principle of entropy is that there will always be a constant amount of order/disorder in a substance. When the magnet puts the substance in 'order', it has to get rid of